BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Combining Programming Constructs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Why must a trace table follow executed assignments in order?

2. Which type suits a count of pupils?

A) Real
B) Boolean
C) Character
D) Integer

3. How do character and string data differ?

4. A limit of 30 appears five times in code. One copy changes to 32; the rest are missed. How would a named constant help? How does it differ from a changing score?

Today's objectives

Combine sequence, selection and iteration to solve problems. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

Sequence

Initialise variables and read inputs before using their values.

Selection

Choose a route after evaluating a Boolean condition.

Iteration

Repeat a block using a loop where a task demands it.

Combined constructs

Real programs often need sequential code, decisions and repetition together.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Sequence	Initialise variables and read inputs before using their values. For example, consider a number guessing game.
Selection	Choose a route after evaluating a Boolean condition. For example, consider a quiz scorer.
Iteration	Repeat a block using a loop where a task demands it. For example, consider a book reminder.
Combined constructs	Real programs often need sequential code, decisions and repetition together. For example, consider a class rewards.

I Do - One worked example

A number guessing game is being developed. Explain how sequence and selection are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Sequence [1]; Initialise variables and read inputs before using their values. [1].

3. Selection [1]; Choose a route after evaluating a Boolean condition. [1].

4. Give one accurate point and a linked reason for each technique.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A quiz scorer is being programmed. Explain why sequence matters. [2 marks]

2. For a book reminder, explain how selection is used. [2 marks]

3. A class rewards needs iteration and combined constructs. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition to print only the even device IDs.

```
for device = 1 to 4
if ... then
print(device)
endif
next device
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how selection and iteration can help a repeat ordering. [4 marks]

Q4. Explain the roles of sequence, iteration and combined constructs when creating a payment kiosk. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Variables should be initialised and required inputs read before their values are used. ☐ T ☐ F
2. Initialise variables and read inputs before using their values ☐ T ☐ F
3. Choose a route after evaluating a Boolean condition ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Each variable value depends on the state produced by earlier executed instructions. ☐

Q2 D - Integer ☐

Q3 A character represents one character; a string represents a sequence of characters. ☐

Q4 Define one named constant, such as MAX_CLASS_SIZE, and use it throughout. Change its definition once to update the limit. Use a variable for a score because it changes while the program runs. ☐

Code First - Model

```
for device = 1 to 4
if device MOD 2 == 0 then
print(device)
endif
next device
```

Output: 2
4

We Do - Check the model

1. Sequence [1]; Initialise variables and read inputs before using their values. [1]. ☐

2. Selection [1]; Choose a route after evaluating a Boolean condition. [1]. ☐

3. Iteration [1]; Repeat a block using a loop where a task demands it. [1]. Combined constructs [1]; Real programs often need sequential code, decisions and repetition together. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Sequence [1]; Iteration [1]. ☐

Q3 Selection [1]; Choose a route after evaluating a Boolean condition. [1]. Iteration [1]; Repeat a block using a loop where a task demands it. [1]. ☐

Q4 Sequence [1]; Initialise variables and read inputs before using their values. [1]. Iteration [1]; Repeat a block using a loop where a task demands it. [1]. Combined constructs [1]; Real programs often need sequential code, decisions and repetition together. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: